Ethical Issues in Supply Chain Management For Demand Forecasting of the Products and Late Deliveries of the Shipment

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***Abstract*—This research report is trying to explore ethical issues in supply chain management industries that is focus on demand forecasting of the products and late deliveries of the shipment. The study is focusing on some questions those are related to future sales and identifying the root cause of the late deliveries and how we can use data analytics to improve performance in supply chain management company. This report is critically identifying ethical issues those are arising in the context of use of the data, personal information available in the dataset, analyzing and hiding the unwanted personal information. It is very important of maintaining confidentiality of the data and we used open source dataset that is available on kaggle which is for general public to use. We have applied all ethical research principles in our throughout study.**

Keywords—Supply chain management, Ethical Issues, Data integrity, accurate information.

I. INTRODUCTION

The efficiency of the supply chain industry significantly influences the performance of the quick delivery of the products and profitability of shipping companies. Now a days Amazon, eBay, Alibaba are very famous for ecommerce industry. They are facing very big challenges e.g. accurate demand forecasting of the products, timely deliveries to the customer and the effective use of the data driven insights is very important for increasing operational efficiency of the ecommerce companies. In this research we are focusing on three main questions which are very helpful for this ecommerce industry. How we can accurately predict future demand based on the previous sale records, It is showing if any particular product sold maximum in last year summer then most probably this summer it will have high demand, secondly most of the customer have complaint about late deliveries of their products. We are trying to find main issues in late deliveries and how we can resolve these problem which arise in late delivery and the utilization of data-driven insights that how we can minimize on time failures. Ethical considerations is very important in all phases of our study as the research involves handling sensitive data of the customers.

A. Objectives of our study:

- We are trying to find the accuracy of the statistical and machine learning models in future demand forecasting based on previous sale.
- Highlight main root causes of late deliveries and how we can tackle these issues.
- Exploring the use of data driven insights that how we can improve supply chain performance.

B. Ethical Considerations:

The ethical considerations are focused in this research for maintaining confidentiality and ensuring the data integrity. We used open source dataset which is available for general public to use. We also focus on ethical concerns and we hide all sensitive information of the dataset during preprocessing step of the implementation. The confidentiality of sensitive personal data is very important to protect from other parties.

II. EVALUATION

A. Ethical Issues Before Starting the Project:

Data Collection and Consent: The dataset for our research is openly available on Kaggle and it is uploaded by DataCo Smart Supply Chain For Big Data Analysis. It is open source dataset which is ensuring that the usage complies with the terms and conditions are specified by the dataset provider. Furthermore, although the data is publicly available so maintaining the confidentiality of the customer sensitive information is always remains a priority. Before starting the project, we verified that whole dataset is used aligns with ethical standards and the legal requirements are very crucial.

B. Ethical Issues During the Research:

Data Handling and Storage:

Managing and storing of large dataset is very challenging especially with a large dataset containing 52 columns and 180,520 rows. During the project we identified missing names and incomplete entries were big challenge. All personal information is not required for the analysis like mobile number address and specific product which is purchased by customer was deleted just to protect individual privacy.

Bias and Fairness in Models:

Selection of appropriate model was crucial for this stage of research. Classification and regression techniques were implemented according to requirements of the problem like regression analysis and support vector machines selected for checking accuracy of the dataset. These models give accurate results which are fulfilling ethical standards and they are error free as compare to other regression and classification models. Evaluating these models are involved checking their performance of different demographic groups just to ensure that no group was excessively get advantaged or disadvantaged.

Transparency and Accountability:

Dataset and its results are totally transparent everyone can get help to reduce late delivery issue and they are accountable because they are available on open source directory. It is essential for creating trust and fulfilling data governance and ethical standards.

C. Ethical Issues After Completing the Research:

Data Anonymization:

Anonymization ensures that the dataset and its results can be utilized by any company or individual for better supply chain management without getting or knowing any personal details of individuals available in the dataset. Only aggregate statistics and results are showing just to ensure that no identifiable information is disclosed.

Impact Assessment:

Our research findings have significant potential for positive impact in supply chain industry. E-commerce and supply chain management companies can use our research findings to overcome issues related to the late deliveries and to forecast future product demand more accurately. Recommendations are giving based on ethical principles just to improve business practices.

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